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Gravity as Computation Time: A Local Budget Rule on a Spin-1/2 Phase-Bit Lattice and its Post-Newtonian Limit

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We ask whether gravity can be modelled as a local limit on how fast each node of a discrete quantum lattice can compute. The test particle is a unitary quantum walker on the face-centred-cubic lattice whose spin is carried by a finite register, the binary octahedral group $2O$: every 60° turn rotates the spinor by a quarter turn about a cube axis. This gives spin 1/2 with Kramers degeneracy at every momentum, the belt-trick sign, and exact particle–antiparticle symmetry; a single extra amplitude, "reverse the heading without rotating the spin", is the lattice Dirac mass $m\beta$ (gap exactly 4μ). Slowing down only the internal clock gives Nordström gravity; a local wait buffer per node gives Einstein's 1911 theory (half light bending). A single node rule—*every process needs each node it involves to be free*, one node for the clock and two for a hop—produces $H = N^2K + NM$, i.e. a spatial metric $A = 1/N$ and $\gamma_{\rm PPN} = 1$ to 3×10^{-4} down to $N = 0.3$. Combining loads at a node as independent occupations ($N = \prod_i N_i$) gives $\beta_{\rm PPN} = 1$ even in the strong field. A direction-independent drag hop supplies g_{0i} with the coefficient that the relativity principle demands. In the full 3+1D walker light is deflected 2.07 times more strongly than with waiting alone, the classical orbits of the lattice metric give the general-relativistic perihelion advance, and a lattice "gravitational atom" built in a slab of the 3D walker reproduces the fine structure $\propto 2\gamma + 1$ (de Sitter) and the Lense–Thirring hyperfine shifts of the Dirac equation in the corresponding metric to 1–4%. We state clearly what is imposed: the field equations for the lapse and the shift, a vacuum counterterm, and the gyroscope rule.

I. INTRODUCTION

In two earlier papers [1, 2] we showed that a path-integral lattice with the single rule "a change of direction costs a factor $i\varepsilon$" reproduces relativistic dispersion, a geometric speed of light, mass, time dilation, and a characteristic internal (crystal angular momentum) structure. The present paper adds gravity.

The guiding idea is simple and old in spirit: if the world is a computation, a region that has more to compute may compute more slowly (Fig. 1). In general relativity (GR) the static weak field is $ds^2 = -N^2dt^2 + A^2d\mathbf{x}^2$ with lapse $N = 1 - U$ and spatial metric $A = 1 + \gamma U$, where $U = GM/rc^2$ and $\gamma = 1$. A lapse alone ($\gamma = 0$) is Einstein's 1911 theory and bends light by half the observed amount [3]. The question is therefore not whether "slower computation" can mimic a redshift—it obviously can—but whether a *local* rule on a lattice also produces the spatial metric, and with it the parameters γ and β of the parametrised post-Newtonian (PPN) formalism [4].

We proceed in three steps. Section II builds the test particle: a unitary 3+1D quantum walker [5, 6] with spin 1/2 from a finite register and a time-reversal-invariant Dirac mass. Section III climbs the ladder from a slowed clock (Nordström) via a local wait buffer (Einstein 1911) to the budget rule ($\gamma = \beta = 1$) and adds the gravitomagnetic shift (overview in Fig. 2). Sections IV and V test the result in the full 3D walker: light deflection, Newton's law, perihelion advance, Lense–Thirring, and the fine and hyperfine structure of a lattice gravitational atom. Section VI separates what follows from the rules from what was put in by hand and lists the lattice-specific signatures.

All results are reproducible with the scripts and data in the public repository [9]; every number quoted below is taken from a stored run.

II. THE PARTICLE: SPIN 1/2 FROM A FINITE PHASE-BIT REGISTER

A. Phase bits

On the hexagonal lattice of Ref. [1] the amplitude of a path is $\prod(i\varepsilon)$ over its turns, i.e. a sum of *integer* path counts weighted by i^T, where T is a $\mathbb{Z}_4$ turn counter. If a turn by $n \cdot 60°$ also advances a phase bit $p \in \mathbb{Z}_4$ by n, only 12 of the 24 labels (d, p) are reachable: the label space is $\mathbb{Z}_{12}$, the double cover of the six headings, and every closed walk of odd winding returns with the sign -1 (checked against exact enumeration up to length 9). In 2D this is the $\alpha = 1/2$ spinor point of the turning-phase model and gives Kramers doublets, but a mass term necessarily breaks time reversal (the 2+1D parity anomaly; we verified that 500 random T-even terms never split the Dirac pair, splitting $\leq 2.9 \times 10^{-15}$).

B. The $2O$ register in 3D

In 3D the holonomy of the minimal SU(2) transport along the cuboctahedron of FCC directions is not quantised: the transports generate an infinite group. We replace each minimal rotation between neighbouring headings (60° apart) by the closest element of the octahedral

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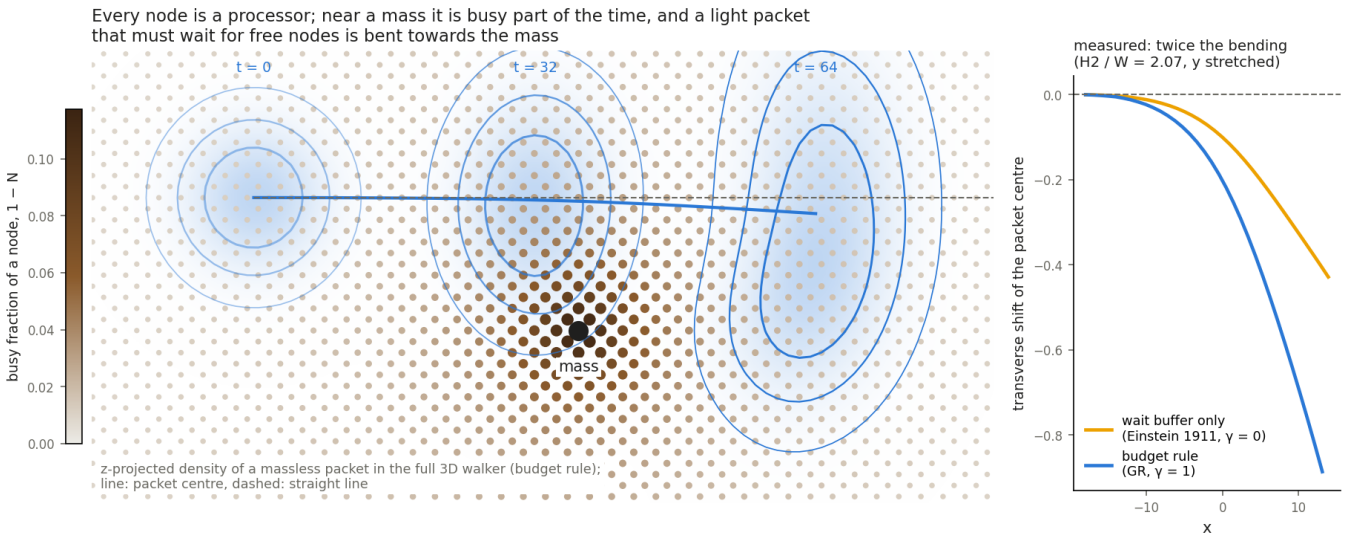


FIG. 1. Light passing a point mass on the phase-bit lattice. Left: the $z = 0$ plane of the FCC lattice; each node is drawn with its busy fraction $1 - N$, $N = e^{-U}$, $U = GM/\sqrt{r^2 + r_c^2}$ ($GM = 0.25$, $r_c = 2$). Blue: z-projected density of a massless packet ($k_0 = 0.6$, $\sigma = 5$, impact parameter $12a$) in the full 3D walker with the budget rule (Sec. III) at three times; the side closer to the mass lags behind and the packet centre (solid line) bends towards the mass. Right: transverse shift of the packet centre with the wait buffer alone and with the budget rule (vertical scale stretched); the deflection ratio is 2.07, the GR value $1 + \gamma = 2$ (Sec. IV).

group. It is unique: the quarter turn about a cube axis,

$$Q(d \to d') = \frac{1 + i\,\mathbf{n}\cdot\boldsymbol{\sigma}}{\sqrt{2}}, \tag{1}$$

with fidelity 0.8165 against 0.5774 for the competing 120° rotation. The 48 matrices Q generate the binary octahedral group $2O$ (the Clifford group of one qubit up to phases). Of the 12×48 conceivable labels (heading, group element) only 48 are reachable, four per heading—again a $\mathbb{Z}_4$ phase bit, now transforming non-abelianly when the heading changes. The integer path-count table, read out with the irreducible representations of $2O$, reproduces the floating-point walk to 1.8×10^{-15} and contains spin 0, $1/2$, 1 and $3/2$ sectors in one table.

The coin is $C = \exp(i\varepsilon G)$ with $G = \sum |d'\rangle\langle d| \otimes Q(d \to d')$. The spin-$1/2$ walker is unitary (9×10^{-16}), causal ($\max|v_g| = 1.687 < c = \sqrt{3}$), Kramers degenerate at every k (6×10^{-15}), and the heading triangle and square carry holonomies π and 2π (all closed walks up to length 8 return with -1). In contrast to the minimal-SU(2) walker the spectrum is exactly symmetric under $E \leftrightarrow -E$ (3.6×10^{-15} against 0.22): the traceless holonomy of the triangles removes odd cycles from the spectral traces.

C. Mass as reversal without spin rotation

The lowest level is a fourfold Dirac point, $H_{\text{eff}} = E_0 + \sum_i k_i \Gamma_i$ with $\{\Gamma_i, \Gamma_j\} = 2c'^2\delta_{ij}$, $c' = c/3$. A local term V opens a gap iff the traceless part of its projection anticommutes with all Γ_i. Scanning all couplings between headings at 60°, 90°, 120°, 180° shows: terms that transport the spin correctly only shift the Dirac point; masses appear exactly where the spin is *not* rotated along with the heading. The simplest is the 180° reversal with identity on the spin, which is the inversion Π itself:

$$C = \exp\big(i(\varepsilon G + \mu\,\Pi)\big), \qquad [\Pi, G] = 0. \tag{2}$$

It is the lattice version of $m\beta$: the gap is exactly 4μ at every order, P, $\mathrm{T} = \Pi\Theta$ and PT are conserved, and with $2O$ the $E \leftrightarrow -E$ symmetry survives (4×10^{-15}). *Mass is the amplitude to reverse the direction of motion without turning the spin.*

The massive quartet carries a common background $-\alpha k^2$ in addition to the Dirac structure. A coupling between headings at 90° with spin transport, $\lambda\varepsilon W_{90}$, commutes with G and cancels it ($\lambda^* = -3.204$ at $\varepsilon = 0.3$); we call the compensated model $2O^*$. At $\varepsilon = 0.3$ its massive band is Dirac to $-0.3\%/-1.0\%$ ($\mu/\varepsilon = 0.1$, $k = 0.2/0.4$) and isotropic. All gravity results below use $2O^*$ at $\varepsilon = 0.3$ with $\Delta t = 1/2$; its limiting speed is $c' = c/3$ and $m = 2\mu$.

III. GRAVITY AS COMPUTATION TIME

A. The vacuum must not gravitate

A lapse rescales quasi-energies. In $2O^*$ the Dirac quartet sits at the Floquet offset $E_c = +1.02$ while the mass is 0.06; without correction the particle would fall 18 times

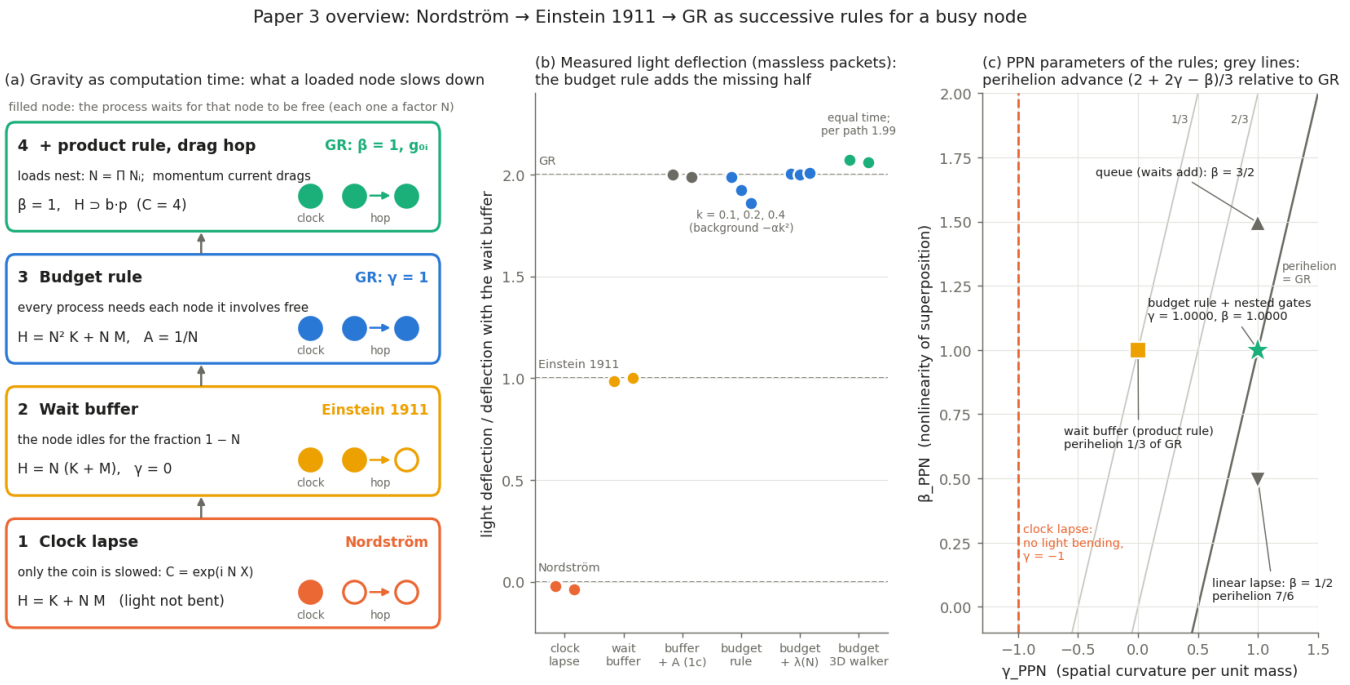


FIG. 2. Overview of the ladder. (a) What a loaded node slows down on each rung; a filled node means the process waits for that node to be free, each contributing a factor N. (b) Measured deflection of massless packets relative to the wait buffer (Einstein 1911): clock lapse and global full lapse on the same chain, local buffer against the global full lapse, Exp. 1c with A set by hand, the budget rule on the chain without and with the $\lambda(N)$ compensation ($k = 0.1, 0.2, 0.4$), and the full 3D walker (equal time). (c) PPN parameters of the rules with lines of constant perihelion advance; only the budget rule with nested gates lies on the GR point.

too strongly. We therefore add a local vacuum counterterm $-\varepsilon\, l_v \mathbb{1}$ that puts the Dirac point at $E = 0$. Then the passive equivalence ratio $R' = E_0/(m^* c'^2)$ is 0.997 for both particle and antiparticle ($\mu/\varepsilon = 0.1$); without the background compensation they would fall differently (0.73 vs 1.20). A residual offset δ gives $R' = 1 \pm \delta/m$, measured on the lattice to three digits—the counterterm is forced, not optional.

B. Clock lapse: Nordström

Slowing only the coin, $C(\mathbf{r}) = \exp(iN(\mathbf{r})X)$, is exactly local and unitary. On a chain of 160 FCC layers with $N = 1 + gx$ resting massive packets fall, moving ones less ($\kappa = (m/E)^2 = 1/\gamma^2$), and massless packets not at all ($a_x = +0.1 \times 10^{-4}$ vs -5.9×10^{-4} at rest). This is Nordström's scalar theory: no light bending.

C. A local wait buffer: Einstein 1911

To slow *everything* locally, each state (x, d, s) receives a buffer partner at the same node:

$$U = R(a)\,\big[\,SC\oplus(-1)\,\big], \qquad R(a) = \cos a + i \sin a\, \sigma_x. \quad (3)$$

The active register makes a normal step, the buffer waits with phase -1, and $R(a)$ mixes the two on the node. Because $R(a)$ is the identity on the internal index, every walker eigenphase θ maps exactly to

$$f_a(\theta) = \tfrac{\theta}{2} + \arcsin\big(\cos a \,\sin\tfrac{\theta}{2}\big) \approx N\theta, \qquad N = \tfrac{1+\cos a}{2}, \quad (4)$$

(checked to 4×10^{-15}). Near $E = 0$ the whole quasi-energy—clock *and* propagation—is scaled by N, and the buffer holds exactly the fraction $1 - N$ of the amplitude: a node with lapse N spends the fraction $1 - N$ of its time waiting. Only the buffer phase π keeps the vacuum at $E = 0$ for every N; any other phase makes the vacuum energy N-dependent, and it would gravitate. On the chain everything now falls alike, light included ($a/(-gN_0^2 c'^2) = 0.86/0.85/0.79$ for massive packets at $k_y = 0/0.3/0.6$, $0.72/0.88$ for massless ones), reproducing the global full lapse $\frac{1}{2}\{N, H\}$: Einstein 1911.

More generally, for *any* local wait scheme the lapse is the active weight of the vacuum fixed vector (Hellmann–Feynman; random two-buffer schemes agree to 10^{-3}), and any combination of wait buffers—also separate ones for hop and coin—scales hop and clock alike: $\gamma = 0$. Waiting alone can never produce a spatial metric.

D. The budget rule: $\gamma = 1$

The clock (coin, mass rotation) involves one node; a hop involves two, departure and arrival. We postulate: *a node is free with weight N, and every process needs all nodes it involves to be free.* Two local gates implement this, a wait gate W (a busy node idles) and a stay gate H (the hop is refused, the coin still acts):

$$\begin{aligned} W(a) &= R(a)\,[\,V \oplus (-1)\,], \\ H(b) &= R(b)\,[\,SC \oplus (-C)\,], \\ U &= W(a)\,\big[\,H(b)\,(SC \oplus -C) \oplus (-1)\,\big]. \end{aligned} \tag{5}$$

With $N_w = N_h = N$ the stay gate slows only the hop, the wait gate slows both, and the band obeys

$$H = N^2 K + NM, \tag{6}$$

with K the massless and M the mass part. This is the weak-field metric with $A = 1/N$. Measured on the band (gap/m and slope/c' extrapolated to $k \to 0$):

N	A	$\gamma = -\ln A/\ln N$
0.95	1.0526	0.9997
0.9	1.1111	0.9998
0.7	1.4285	0.9999
0.5	2.0000	1.0000
0.3	3.3333	1.0000

So $g_{00}g_{rr} = -1$ holds into the strong field. The general count is $\gamma = (n_{\rm hop} - n_{\rm clock})/n_{\rm clock}$; for one and two nodes it is exactly 1, and since every hop in every direction involves two nodes the metric is isotropic. In Bloch-sphere language each gate is a register qubit whose fixed vector has polar angle a; the lapse is its active weight $\cos^2(a/2)$.

Price and repair. The stay gate adds a band curvature $-\alpha k^2$, $\alpha = 0.245\,N_h(1-N_h)$, of order Δt. Because λ changes only this background (Dirac point, gap and velocity fixed to 2×10^{-4}), a node that retunes $\lambda(N)$ ($\Delta\lambda \approx 3.3(1-N)$) removes it for $N \gtrsim 0.4$. On the chain light is then deflected twice as strongly as with waiting alone, $(a_{H2}/N_0^2)/a_{1b} = 2.005/1.998/2.007$ at $k = 0.1/0.2/0.4$, i.e. $\gamma = 1.00 \pm 0.007$; massive Dirac-regime packets give $\gamma = 0.99$–1.00 (Fig. 3).

E. Combining loads: $\beta = 1$

In isotropic PPN gauge $\ln N = -U + (\beta-1)U^2$. If the field law is linear in the sources, β is fixed by how two loads combine at one node:

$$\beta - 1 = \frac{\ln N_{12} - \ln N_1 - \ln N_2}{2\ln N_1 \ln N_2}. \tag{7}$$

We measured three composition laws on the band:

law	gate	β
linear, $N = 1 - \sum U_i$	one	0.20–0.50
queue, $N^{-1} - 1 = \sum(N_i^{-1} - 1)$	parallel	1.37–1.50
product, $N = \prod N_i$	**nested**	**1.0000**

Nested gates—a state must be free for *each* load—give $\beta = 1$ to 10^{-4} up to $N_{12} = 0.49$, and keep $A = 1/(N_1N_2)$. This is the same logic as the budget rule itself (a hop needs $N \cdot N$, a state needs $N_1 \cdot N_2$). The field must therefore couple as $N = N_0 e^{\Phi}$, not $N_0(1+\Phi)$. Queues and linear superposition are excluded by the perihelion of Mercury and lunar laser ranging [4].

F. The source: computational load

The source of Φ is the local load $e(x) = \mathrm{Re}\,\psi^\dagger i(U\psi - \psi)/\Delta t$ (zero in the vacuum) and the mass load $e_M = \mu\,\mathrm{Re}[i(U\psi)^\dagger \partial_\mu U\psi]/\Delta t$, both local. The combination $s = 2e - e_M$ equals $T_{00} + T_{ii}$: a moving source pulls with $1.887/1.964$ against $1+v^2 = 1.892/1.970$, light with 2. The mass load is the proper time, $e_M/m = d\tau/dt = N\sqrt{1-v^2}$ (lattice band to 0.9%, 40 cases), and its space-time integral is $m\tau = -S$. A resting source in a potential loads with Nm (2×10^{-5}); together with the product rule this gives

$$g_{00} = -1 + 2U^* - 2U^{*2} + 3\Phi_1^* - 2\Phi_2^* \tag{8}$$

in the conserved density ρ^*, which is the 1PN g_{00} of GR. For antimatter the load of the lower band is negative; only the hole picture (load relative to the filled sea) gives normal attraction ($\pm1.858\times10^{-4}$ for a particle–antiparticle pair), at the price of normal ordering the sea. With a local wave equation $\Phi_{tt} = c_g^2(\nabla^2\Phi - \kappa s)$, $c_g = c'$, a test particle starts to fall only after the light travel time; in 3D the static field is $1/r$ to 0–2% and does not grow secularly.

G. Gravitomagnetism: the drag hop

With scalars N and A only, the lattice would be a preferred frame for gravity. For two sheets with speeds v_s, v_t the transverse force carries the factor $(1+v_s^2)(1+v_t^2) - Cv_sv_t$, and co-moving bodies cannot detect their motion ($a \propto 1/\gamma^3$) only if $C = 4$, the GR value. A heading-dependent stay gate gives a linear term but splits Kramers pairs and shifts the vacuum: in this walker the internal index *is* the direction of motion. The working solution is a *direction-independent drag hop*: every node hands a small fraction of every state, whatever its heading, to its neighbour along the momentum current. It generates $H \supset \mathbf{b}\cdot\mathbf{p}$ exactly (5×10^{-15} on the band), leaves the Dirac point and Kramers pairs untouched, and with $\mathbf{b} \propto 4c'^2\kappa\nabla^{-2}$(momentum density) gives on a 640-layer chain: parallel light beams 0.019 (packet GR 0.022; without g_{0i} 0.053), antiparallel 1.99, co-moving sheets $a(k,k)/a(0,0) = 0.669/0.319$ against $0.675/0.339$ (without g_{0i}: $1.77/3.63$). Causally propagated (Φ and $\mathbf{b}$ as

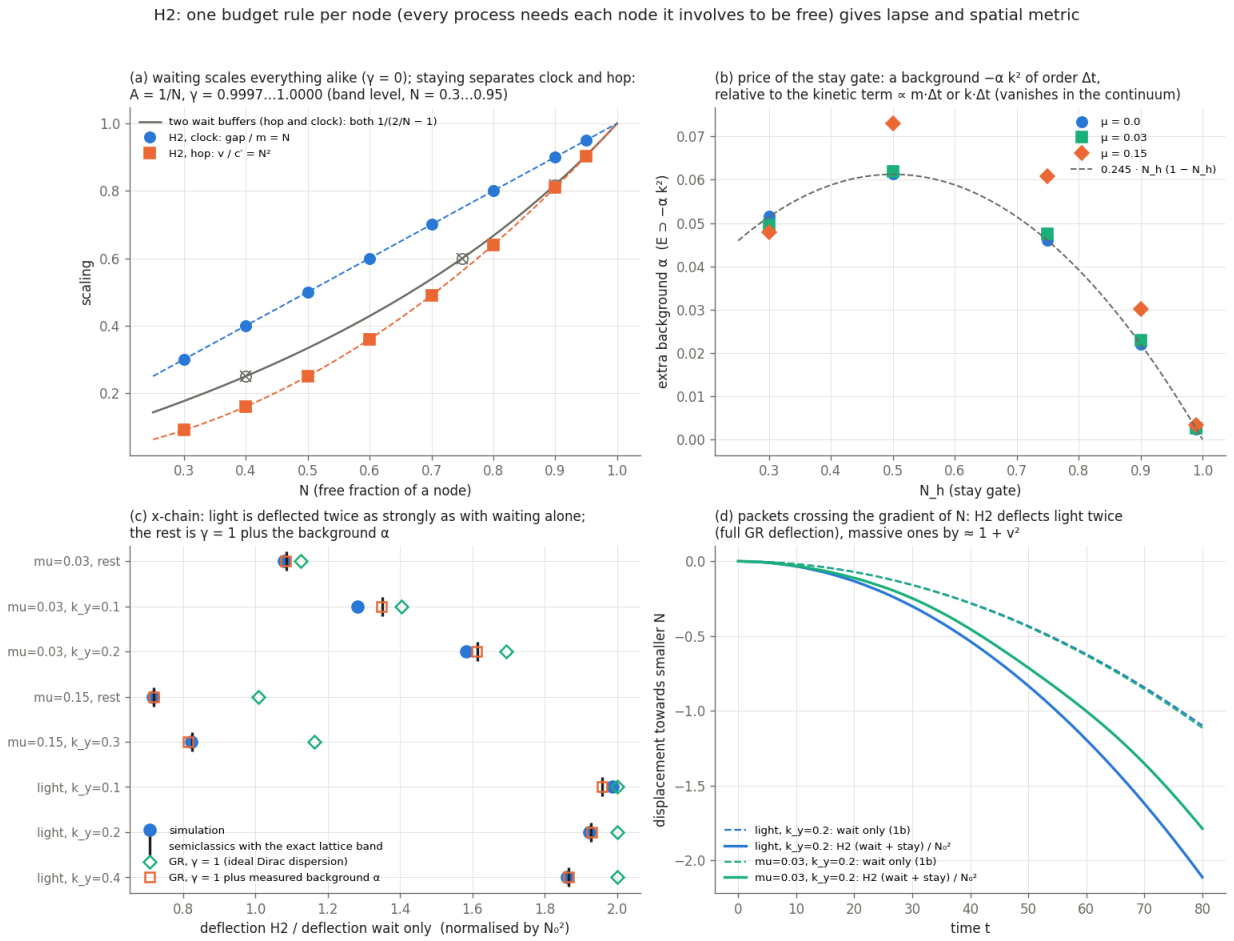


FIG. 3. The budget rule (H2). Waiting alone is always a pure lapse; the stay gate slows only hops, giving $H = N^2K + NM$, $A = 1/N$ and $\gamma_{\rm PPN} = 1$; on an x-chain light is deflected twice as strongly as with the wait buffer (Einstein 1911).

wave equations), parallel light beams feel nothing at any time; an instantaneous $\mathbf{b}$ next to a retarded Φ would give a superluminal repulsion.

In 3D the drag is a brick wall of real bond rotations between FCC neighbours, angle $\Delta t\,(\mathbf{b}\cdot\mathbf{s})/(8\ell^2)$ at the bond midpoint, exactly unitary and strictly local. A rotating ball gives the GR dipole $\mathbf{b} = 2c'^2(\kappa/4\pi)\,\mathbf{J}\times\mathbf{r}/r^3$ (0.99 at $r = 8$). In a rigidly rotating drag field a packet is carried along (0.00406 against $\Omega = 0.004$), but its spin is not (0.00026): $\mathbf{b}\cdot\mathbf{p} = \Omega L_z$ couples to the orbit only. A gyroscope needs the additional rule $\exp\!\big(-i\Delta t\,\frac{1}{2}\nabla\times\mathbf{b}\cdot J_{\rm int}\big)$ with the lattice's own rotation generator $J_{\rm int} = (2i/\pi)\log R_{90}$ (R_{90}: heading permutation times spin rotation, commutes with the coin). It has $g = 1$ on every spin-½ doublet, and the spin then turns with 0.00399. The bare spin index $P(\mathbb{1}\otimes\boldsymbol{\sigma}/2)P$ would give $g = -1/3$ for the particle: most of its spin sits in the headings.

IV. TESTS IN THE FULL 3+1D WALKER

A. Light deflection by a point mass

The full 3D walker runs on a periodic $112\times88\times80$ box (7.9×10^5 nodes, four registers, complex64), with $U = GM/\sqrt{r^2+r_c^2}$, $GM = 0.25$, $r_c = 2$, $N = e^{-U}$, and coins tabulated in 48 classes of N. A massless packet ($k_0 = 0.6$, $\sigma = 5$) passes the mass at impact parameter b. Both Kramers partners give the same deflection to 10^{-5}.

b	rule	Δk_y sim.	sim./local force	H2/W
7.35	W	−0.02174	1.018	
7.35	H2	−0.04503	1.027	**2.071**
12.25	W	−0.01907	1.000	
12.25	H2	−0.03925	1.012	**2.058**

The budget rule doubles the deflection in 3D: $4GM/b$ against Einstein's $2GM/b$. The excess over 2 is a time effect (light is slower in H2, $c'N^2$, and spends longer near the mass); per distance travelled the straight-ray estimate

is 1.99/1.98. The local force $n_s N^{n_s} c' \langle |k| \rangle \partial_y U$ ($n_s = 1$ for W, 2 for H2), averaged over the simulated density, matches to $\leq 2.7\%$ (Fig. 4).

B. Newton's law and the perihelion

A resting massive packet ($m = 0.3$, $\sigma = 5$) in the field of a point mass follows the local force of the metric, $-\langle \partial_x N \sqrt{m^2 + (N^n c' k)^2} \rangle$, to $\leq 1\%$ for $r \geq 13$; the force lies 8% (W) and 28% (H2) above mGM/r^2 because the packet's momentum spread ($v^2 \approx 0.18$) gravitates too. A wave packet on a classical orbit is out of reach ($L \approx 1\hbar$, spreading much faster than one period), so the perihelion is computed from the classical orbits ($a = 1600$) of the walker's effective Hamiltonian $H = N\sqrt{m^2 + c'^2 p^2 / A^2}$:

variant	(γ, β)	/PPN	/GR
H2, $A = 1/N$	$(1, 1)$	1.0026	1.00
wait only, $A = 1$	$(0, 1)$	1.0011	1/3
linear lapse	$(1, 1/2)$	1.0039	7/6

The PPN formula $(2 + 2\gamma - \beta)/3$ holds for all three; the budget rule with the product law gives the full GR advance. A classical orbit with $H \supset \mathbf{b} \cdot \mathbf{p}$ precesses its node at the Lense–Thirring rate to 1.001.

C. What an observer inside the lattice sees

An observer defines spheres by their own signals (radar time). The cuboctahedral graph shells are invisible to them—no signal follows graph distance; the weight of a point source at graph distance $\geq 0.8n$ falls from 0.021 at $n = 8$ to below 10^{-3} for $n \geq 32$. What they can test is agreement between probes. Over 2000 directions the speed of the massless doublet varies by 1.1×10^{-4} at $k = 0.05$ ($\propto k^{1.83}$, a $k^3 Y_4$ term) and depends on energy as $v = c'(1 - \xi(ka)^2)$, $\xi = 0.42 \pm 0.05$. Michelson–Morley with long-wavelength light finds nothing to order $(ka)^2$; the particle's own isotropy improves as $C_4 \propto \sigma^{-4}$ with packet width.

V. THE GRAVITATIONAL ATOM

A bound state is the sharpest test of a metric: level shifts are measured as frequencies, to many digits. For a light particle the "gravitational atom" has $L \sim \hbar$, so it is the quantum counterpart of the perihelion problem.

A. Effective Hamiltonian and l-splitting

On a 1D chain of 512 layers ($\mu = 0.08$, $GM = 0.6$, 60 000 ticks) the lattice levels agree with the effective Hamiltonian $H = N^{1/2}\sqrt{m^2 + c'^2 \Pi^2}\, N^{1/2}$, $\Pi^2 = N^{n/2}(-\nabla N^n \nabla) N^{n/2}$, to 0.2% (ground) to 3%, including the difference between H2 and W. In 3D (radial, effective) the relativistic l-splitting is the quantum perihelion: wait-only/H2 tends to $0.22 \to 0.31$ (classically 1/3), linear/H2 to 1.16–1.18 (7/6), scaling as $(GM)^4$.

B. Fine structure: the de Sitter factor $2\gamma + 1$

For a Dirac particle in $ds^2 = -N^2 dt^2 + A^2 d\mathbf{x}^2$, $H = \beta m N + \frac{c'}{2}\{N/A, \boldsymbol{\alpha} \cdot \mathbf{p}\}$, the Foldy–Wouthuysen expansion contains

$$H_{\mathrm{SO}} = (2\gamma + 1)\, \frac{c'^2}{4m}\, \frac{1}{r}\, \frac{dU}{dr}\, \boldsymbol{\sigma} \cdot \mathbf{L}, \tag{9}$$

i.e. $\gamma + 1/2$ per unit of spin (de Sitter precession 3/2 for $\gamma = 1$, the Hehl–Ni term 1/2 for $\gamma = 0$ [7]). Hence $E(2p_{3/2}) - E(2p_{1/2})$ is in the ratio 1 : 3 : 3 for wait-only : H2 : linear lapse, which the radial 2D Dirac reference confirms at weak coupling (1 : 2.997 : 3.011) and turns into 1 : 3.64 : 4.50 at $\alpha_f = m\,GM/c' = 0.3$.

A 3D atom does not fit into memory, but a 2D atom in a slab of the full 3D walker does, and suffices because spin and orbital angular momentum are both normal to the slab. We run `Walker3D` on $340 \times 340 \times 2$ ($\mu = 0.1$, $\alpha_f = 0.3$, Bohr radius $a_0 = 9.62$, $r_c = a_0/2$, 8000 ticks), start from the reference $2p$ shell times $e^{i\varphi}$ times a rest spinor with $J_z = \pm 1/2$, project exactly onto the C_4 sector, and extract the level by filter diagonalisation of the autocorrelation [8] (the FFT resolution is 4–15 times too coarse). The reference includes the spin-independent Wilson-like term of the lattice band ($r_b = -0.037$, measured from the band, not fitted):

rule	lattice (10^{-4})	Dirac + r_b	ratio
wait only ($\gamma = 0$)	1.0715	1.0881	0.985
H2 ($\gamma = \beta = 1$)	3.9928	3.9621	1.008
linear ($\gamma = 1, \beta = 1/2$)	4.9590	4.9009	1.012

The walker has the gravitational spin–orbit coupling with the right size and sign ($2p_{3/2}$ above $2p_{1/2}$), and its dependence on spatial curvature: H2/W = 3.73 against 3.64. Absolute $2p$ levels agree to 0.3–1.1% (Fig. 5).

C. Hyperfine structure: Lense–Thirring

A rotating central body adds $\mathbf{b} = B_J\, \hat{\mathbf{z}} \times \mathbf{r}/(r^2 + r_{c,b}^2)^{3/2}$. The walker couples through the drag hop ($\mathbf{b} \cdot \mathbf{p} = (b_\varphi/\rho) L_z$) and the gyroscope rule ($\frac{1}{2} \nabla \times \mathbf{b} \cdot J_{\mathrm{int}}$). Far from the core, orbit : spin = 2 : −1 as in GR ($2GJ \cdot L/c^2 r^3$ and $\Omega_{\mathrm{LT}} = -GJ/c^2 r^3$ in the equatorial plane), so the shifts are $\propto m_l - s_z/2$ and $2p_{3/2} : 2p_{1/2} = 3 : 5$; without spin coupling 1 : 1, with $g = 2$, 1 : 3. The ratio measures the gyrogravitomagnetic factor. With the core ($r_{c,b} = 0.25 a_0$) and strong field the Dirac reference gives 0.560. Six lattice runs ($\pm J$, with and without the spin rule, 4.9 h each):

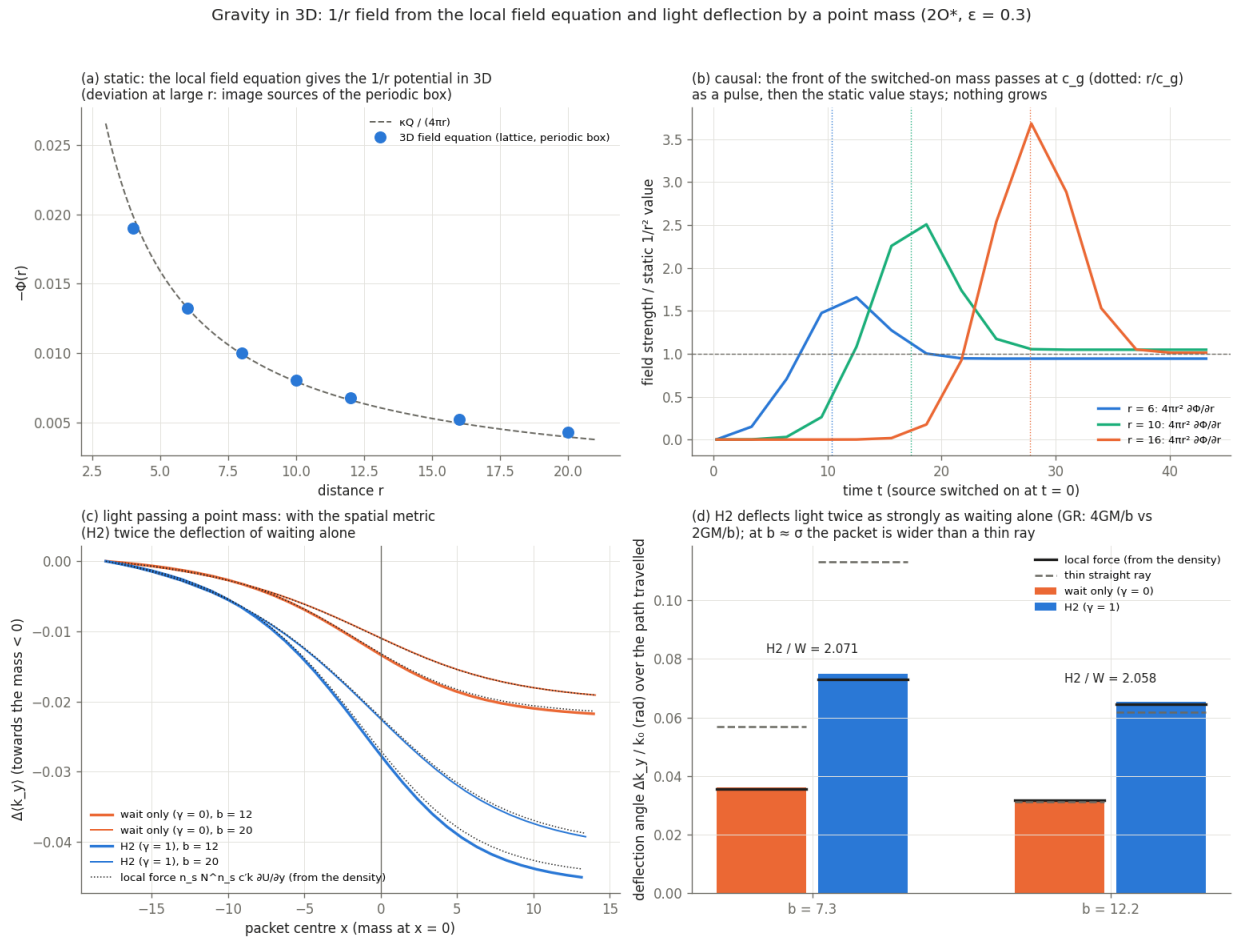


FIG. 4. Gravity in 3D: static and causal $1/r$ field, and light deflection by a point mass in the full $112 \times 88 \times 80$ walker; the budget rule (H2) deflects 2.07 times more strongly than waiting alone (W).

shift (10^{-5})	lattice	reference	ratio
$2p_{3/2}$, orbit only	4.808	4.678	1.028
$2p_{1/2}$, orbit only	6.357	6.187	1.027
$2p_{3/2}$, orbit + spin	4.295	4.151	1.035
$2p_{1/2}$, orbit + spin	7.618	7.407	1.028
spin part $2p_{3/2}$	−0.646	−0.659	0.980
spin part $2p_{1/2}$	+1.077	+1.036	1.039

The ratio $2p_{3/2} : 2p_{1/2}$ is 0.5638 on the lattice against 0.5604 (orbit only: 0.7564 vs 0.7560), and even the part quadratic in J agrees to 4–6%. The common factor 1.03 of all orbital shifts is spin-independent; it is either the slightly more compact lattice shell or the drag hop itself (+1.5% in the rigid-rotation test), not yet separated.

VI. DISCUSSION

A. What follows from the rules, and what is imposed

Derived from local rules: spin $^1/_2$, Kramers and the belt-trick sign from a finite register; the Dirac mass as reversal without spin rotation; the lapse as the waiting fraction of a node; $\gamma = 1$ from counting the nodes a process involves; $\beta = 1$ from combining loads as independent occupations; the source $T_{00} + T_{ii}$ and proper time from the computational load; the coefficient $C = 4$ of the drag from the relativity principle; and the quantitative agreement of all bound-state tests with the Dirac equation in the resulting metric.

Imposed: (i) the field equations for Φ and $\mathbf{b}$ (Poisson or wave equation with the measured sources)—the budget rule says how a node reacts to its load, not how the load spreads; (ii) the vacuum counterterm and the $\lambda(N)$ retuning; (iii) the gyroscope rule $\frac{1}{2}\nabla \times \mathbf{b} \cdot J_{\rm int}$, which the lattice does not enforce (the bare spin index gives $g = -1/3$); (iv) the equal weighting of departure and

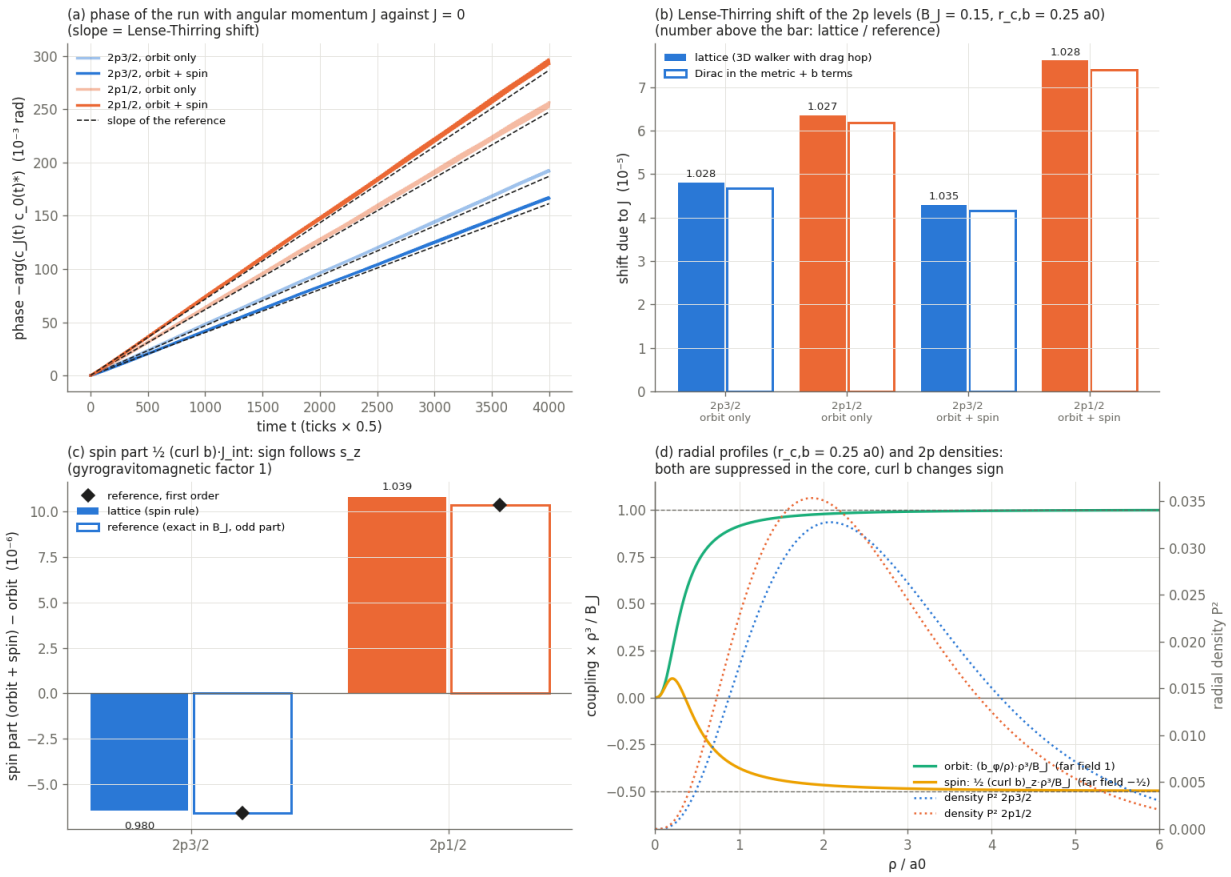


FIG. 5. Gravitational hyperfine structure. (a) Phase of the run with angular momentum J against $J = 0$; the slope is the Lense–Thirring shift. (b) Shifts of the $2p$ levels, lattice against the Dirac equation with gravitomagnetic terms. (c) Spin part: its sign follows s_z (gyrogravitomagnetic factor 1). (d) Radial profiles of the orbital and spin couplings with the $2p$ densities.

arrival nodes ($N_w = N_h$).

B. Lattice signatures

The model is not GR on a grid; it has its own predictions:

- *Mirror instead of horizon.* Approaching $N \to 0$ a photon is blue-shifted up to the band maximum at $k = \pi/a$ and then turns back ($N = 0.089$ in our run); massive particles bounce back almost to their starting height.
- *Energy-dependent light speed* $1 - 0.42(ka)^2$ and a cubic anisotropy $\propto (ka)^2$; quadratic in a, this is compatible with gamma-ray-burst timing if a is near the Planck length.
- *Vacuum energy.* The Floquet offset gravitates unless subtracted, and the Dirac sea must be normal ordered—the cosmological-constant problem appears explicitly as a lattice counterterm.
- *Background of order* Δt from the stay gate, which breaks local Lorentz invariance at finite k in loaded regions unless compensated.

C. Open problems

The decisive open question is the *tensor sector*. The model has two scalars (N, $A = 1/N$) and a vector ($\mathbf{b}$); GR also has transverse-traceless radiation. Many theories reproduce $\gamma = \beta = 1$ in the static weak field and fail on gravitational waves. Whether an oscillating load on the lattice radiates two tensor polarisations, or whether the budget rule is a scalar–vector theory that only mimics GR in the solar system, is the next test. Equally important is a derivation of the gyroscope rule from a drag hop that also rotates the $2O$ register, and a field equation for Φ and $\mathbf{b}$ that follows from the node rule rather than being added to it.

VII. CONCLUSION

A unitary walker with a finite spin register and one local node rule—every process needs each node it involves to be free—reproduces the post-Newtonian limit of general relativity with $\gamma = \beta = 1$, full light bending in 3+1D, the GR perihelion advance, the Lense–Thirring dipole, and the fine and hyperfine structure of a gravitational atom to a few percent. Nordström, Einstein 1911 and GR appear as successive rungs: slow the clock, slow everything, slow hops twice. Gravity, in this model, is the bookkeeping of which nodes are busy.

ACKNOWLEDGMENTS

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